MATH 350 Advanced Calculus

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August 26, 2026

Outline

1 Real Analysis Lecture 2
- Types of real numbers
- The tenth axiom

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   - Types of real numbers
   - The tenth axiom

The problem with axioms

Problems when axiomatizing:

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- intuitive axioms

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We need to redetermine basics

For us, we've lost some basic important things...

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- what is 3?

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- We have 0 and 1 by axioms
- what is 2?

  Definition: $2 = 1 + 1$

- what is 3?

  Definition: $3 = 2 + 1$

- are 1, 2, 3 all different?

Challenge!

Problem

Prove that 1, 2, and 3 are all different.

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Solution

Challenge!

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Solution

Since $0 < 1$, A7 says $0 + 1 < 1 + 1$.

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Since $0 < 1$, A7 says $0 + 1 < 1 + 1$.
Therefore $1 < 2$.

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Since $0 < 1$, A7 says $0 + 1 < 1 + 1$.
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Then again by A7, $1 + 1 < 2 + 1$.

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Solution

Since $0 < 1$, A7 says $0 + 1 < 1 + 1$.
Therefore $1 < 2$.
Then again by A7, $1 + 1 < 2 + 1$.
Therefore $2 < 3$.
Also transitivity (A9) says $1 < 2$ and $2 < 3$ implies $1 < 3$.

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Problem

Prove that 1, 2, and 3 are all different.

Solution

Since $0 < 1$, A7 says $0 + 1 < 1 + 1$.
Therefore $1 < 2$.
Then again by A7, $1 + 1 < 2 + 1$.
Therefore $2 < 3$.
Also transitivity (A9) says $1 < 2$ and $2 < 3$ implies $1 < 3$.
The trichotomy then implies $1 \neq 2$ and $2 \neq 3$ and $1 \neq 3$.

Natural numbers

Question

What are the natural numbers $1, 2, 3, \ldots$ based on the Axioms?

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- $1 \in \mathbb{N}$

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Natural numbers

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What are the natural numbers $1, 2, 3, \ldots$ based on the Axioms?

Definition

- a set $\mathbb{N}$
- $1 \in \mathbb{N}$
- $2 \in \mathbb{N}$
- $3 \in \mathbb{N}$
- general property:

$$x \in \mathbb{N} \implies x + 1 \in \mathbb{N}.$$

Inductive sets

Definition

A subset S of $\mathbb{R}$ is called a **inductive set** if

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Is there more than one inductive set?

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- $1 \in S$
- if $x \in S$ then $x + 1 \in S$

Question

Is there more than one inductive set?

$$\mathbb{R}, \quad \mathbb{Q}, \quad (0, \infty), \quad \mathbb{Z}, \quad \mathbb{N}, \ldots$$

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$$\mathbb{Z} = \{x : x \text{ is zero or } \pm x \text{ is a positive integer}\}$$

- rationals are

$$\mathbb{Q} = \left\{\frac{a}{b} : a, b \in \mathbb{Z},\ b \neq 0\right\}.$$

Induction

Principle of Induction:

If S is an inductive set, then $\mathbb{Z}_+ \subseteq S$

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Example: Let's prove $n(n+1)$ is divisible by 2 for all positive integers n, using the Principle of Induction.

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Proof:

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Example: Let's prove $n(n+1)$ is divisible by 2 for all positive integers n, using the Principle of Induction.

Proof:

Let

$$S = \{n \in \mathbb{Z}_+ : 2|n(n+1)\}$$

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. We see that $1 \in S$ because 2 divides $1(1+1)$.

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Now suppose $x \in S$. (This is our usual inductive assumption).

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Then 2 divides $x(x+1)$, so $x(x+1) = 2k$ for some integer k.

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Then 2 divides $x(x+1)$, so $x(x+1) = 2k$ for some integer k.
This means $(x+1)(x+2) = x(x+1) + 2(x+1) = 2(k+x+1)$.

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Thus 2 divides $(x+1)(x+2)$, showing that $x+1 \in S$.

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By the Principle of Induction, this means $\mathbb{Z}_+ \subseteq S$.

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In other words $n \in S$ for every positive integer n.

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By the Principle of Induction, this means $\mathbb{Z}_+ \subseteq S$.
In other words $n \in S$ for every positive integer n.
Hence 2 divides $n(n+1)$ for every positive integers n.

Other numbers

Question

What about other kinds of numbers, like irrational algebraic numbers or transcendentals? Why do they exist according to the axioms?

To answer this, we need to understand the tenth axiom of the real numbers:

the completeness axiom

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Real numbers

The **real numbers** are the unique field $\mathbb{R}$ satisfying Axiom 1-Axiom 9, plus an extra axiom called the completeness axiom.

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- A10 **completeness axiom**: every nonempty set of numbers $S \subseteq F$ which is bounded above has a supremum $\sup(S)$

- Understanding what this means is challenging, but will be fundamental to the entire course!
- Intuitively, it represents the fact that the real line has no holes or gaps.

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Examples:

- 34 is an upper bound of $[-1, 5]$, but not a maximal element
- 5 is a maximal element of $[-1, 5]$
- 3 is an upper bound of $[0, 3)$, but not a maximal element
- $\mathbb{Z}_+$ has no upper bound
- $[3, 7)$ has an upper bound but no maximal element

Challenge!

Problem

Show that if S has a maximal element, then it is unique.

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Hint: use the definition!

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Solution

Suppose that b_1 and b_2 are both maximal elements of S.

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Solution

Suppose that b_1 and b_2 are both maximal elements of S.
Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.

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Solution

Suppose that b_1 and b_2 are both maximal elements of S.
Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.
Moreover, $b_1 \in S$ and $b_2 \in S$.

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Suppose that b_1 and b_2 are both maximal elements of S.
Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.
Moreover, $b_1 \in S$ and $b_2 \in S$.
Since $b_1 \in S$ and b_2 is an upper bound of S, $b_1 \leq b_2$.

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Show that if S has a maximal element, then it is unique.

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Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.
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Since $b_1 \in S$ and b_2 is an upper bound of S, $b_1 \leq b_2$.
Likewise, since $b_2 \in S$ and b_1 is an upper bound of S, $b_2 \leq b_1$.

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Likewise, since $b_2 \in S$ and b_1 is an upper bound of S, $b_2 \leq b_1$.
By the trichotomy, we find $b_1 = b_2$.

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Suppose that b_1 and b_2 are both maximal elements of S.
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Since $b_1 \in S$ and b_2 is an upper bound of S, $b_1 \leq b_2$.
Likewise, since $b_2 \in S$ and b_1 is an upper bound of S, $b_2 \leq b_1$.
By the trichotomy, we find $b_1 = b_2$.

Now we can say *the* maximum, $\max(S)$

Challenge

Problem

Prove that if A is a nonempty set of integers that is bounded above, then A has a maximum.

Challenge

Solution

The set A is bounded above, so it has a supremum $b = \sup(A)$.

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The set A is bounded above, so it has a supremum $b = \sup(A)$. Since $b - 1$ is not an upper bound, so there exists $a \in A$ with $b - 1 < a$.

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If $a' \in A$, then a' is an integer and therefore $a' = a + k$ for $k \in \mathbb{Z}$.

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Solution

The set A is bounded above, so it has a supremum $b = \sup(A)$.
Since $b - 1$ is not an upper bound, so there exists $a \in A$ with $b - 1 < a$.
If $a' \in A$, then a' is an integer and therefore $a' = a + k$ for $k \in \mathbb{Z}$.
Since b is an upper bound, $b \geq a + k > b - 1 + k$, making $0 > k - 1$.

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Solution

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Since $b - 1$ is not an upper bound, so there exists $a \in A$ with $b - 1 < a$.
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Therefore $k \leq 0$ and $a' \leq a$.

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Solution

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Since $b - 1$ is not an upper bound, so there exists $a \in A$ with $b - 1 < a$.
If $a' \in A$, then a' is an integer and therefore $a' = a + k$ for $k \in \mathbb{Z}$.
Since b is an upper bound, $b \geq a + k > b - 1 + k$, making $0 > k - 1$.
Therefore $k \leq 0$ and $a' \leq a$.
It follows that a is an upper bound of A, and since $a \in A$ it is a maximum.

Supremum

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- if $b' < b$, then b' is not an upper bound of S

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- if $b' < b$, then b' is not an upper bound of S

In other words

a supremum is a **least upper bound**

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Solution

Suppose that b_1 and b_2 are both suprema of S.

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Show that if S has a supremum, then it is unique.

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Solution

Suppose that b_1 and b_2 are both suprema of S.
Then b_1 and b_2 are both upper bounds of S.

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Solution

Suppose that b_1 and b_2 are both suprema of S.
Then b_1 and b_2 are both upper bounds of S.
Since b_1 is a least upper bound, $b_1 \leq b_2$.

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Suppose that b_1 and b_2 are both suprema of S.
Then b_1 and b_2 are both upper bounds of S.
Since b_1 is a least upper bound, $b_1 \leq b_2$.
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Suppose that b_1 and b_2 are both suprema of S.
Then b_1 and b_2 are both upper bounds of S.
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Since b_2 is also a least upper bound, $b_2 \leq b_1$.
By the trichotomy, we find $b_1 = b_2$.

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Now we can say *the* supremum, $\sup(S)$

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Show that if S has a maximum element. Then S has a supremum and $\max(S) = \sup(S)$

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Solution

Let $b = \max(S)$.

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Solution

Let $b = \max(S)$.
Then b is an upper bound of S and $b \in S$.

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Show that if S has a maximum element. Then S has a supremum and $\max(S) = \sup(S)$

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Solution

Let $b = \max(S)$.
Then b is an upper bound of S and $b \in S$.
If b' is another upper bound of S, then by definition $b \leq b'$.

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Show that if S has a maximum element. Then S has a supremum and $\max(S) = \sup(S)$

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Solution

Let $b = \max(S)$.
Then b is an upper bound of S and $b \in S$.
If b' is another upper bound of S, then by definition $b \leq b'$.
Thus b is the least upper bound of S.

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- 2 is the supremum of

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- 1 is the supremum of

$$\{\sin(n) : n \in \mathbb{Z}_+\}$$

Decimals

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Definition

The k'**th decimal apprixomation** of $n.d_0 d_1 d_2 d_3 \dots$ is

$$s_k = n + \frac{d_1}{10} + \frac{d_2}{10^2} + \frac{d_3}{10^3} + \cdots + \frac{d_k}{10^k}.$$

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The **decimal** $n.d_0 d_1 d_2 d_3 \ldots$ is the real number defined as the supremum of decimal approximations

$$n.d_0 d_1 d_2 d_3 \cdots = \sup\{s_k : k \in \mathbb{N} \ldots\}.$$

Examples of decimals

- Liouville's constant

$$0.11000100000000000000000010\ldots$$

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- later we will show *any* positive number has at least one decimal expansion

Lower bounds and infima

A **lower bound** for a set $S \subseteq \mathbb{R}$ is a number b such that

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In other words

an infimum is a **greatest lower bound**

Properties of Suprema

The first property of suprema is that they must be arbitrarily close to elements of the set.

Theorem (Approximation Property)

Let $S \subseteq \mathbb{R}$ be nonempty and bounded above, and let $b = \sup(S)$. Then for all $\epsilon > 0$, there exists $a \in S$ with

$$b - \epsilon < a \leq b.$$

Properties of Suprema

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Therefore thre must exist $a \in S$ with $a > b - \epsilon$. □

Properties of Suprema

Also, suprema play nicely with addition.

Theorem (Additive Property)

Let $A, B \subseteq \mathbb{R}$ be nonempty, bounded above sets.

$$C = \{x + y : x \in A,\ y \in B\}.$$

Then C is bounded above and

$$\sup(C) = \sup(A) + \sup(B).$$

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If $z \in C$, then $z = x + y$ for some $x \in A$ and $y \in B$, and therefore $z = x + y \leq a + b$.

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Thus $c - a$ is an upper bound of B, and it follows $b \leq c - a$.
Therefore $a + b \leq c$. $\square$

Challenge

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Use the completeness axiom to prove that the $\mathbb{Z}_+$ is not bounded above. (Apostol Theorem 1.17)

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Then $b - 1 < b$ by Axiom 7, so $b - 1$ cannot be an upper bound.

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It follows from Axiom 7 that $b < n + 1$.

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This means there exists $n \in \mathbb{Z}_+$ with $b - 1 < n$.
It follows from Axiom 7 that $b < n + 1$.
However, $n + 1 \in \mathbb{Z}$, so this contradicts b being an upper bound.

Archimedian property

Theorem (Apostol Theorem 1.18)

For every $x \in \mathbb{R}$, there exists $n \in \mathbb{Z}_+$ with $n > x$.

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Theorem (Archimedian Property of Reals)

For every $x, y \in \mathbb{R}$ with $x > 0$, there exists $n \in \mathbb{Z}_+$ with $y < nx$.

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Proof.

Replace x with y/x in the previous theorem. □